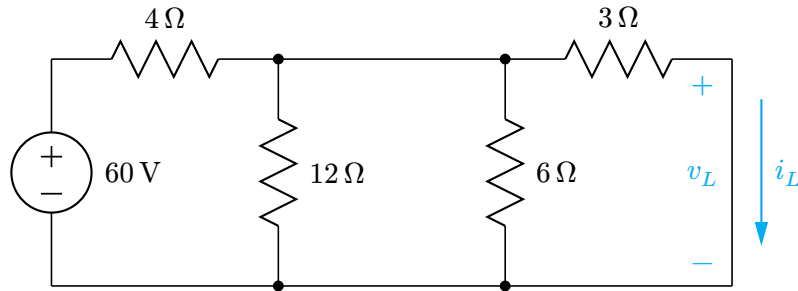


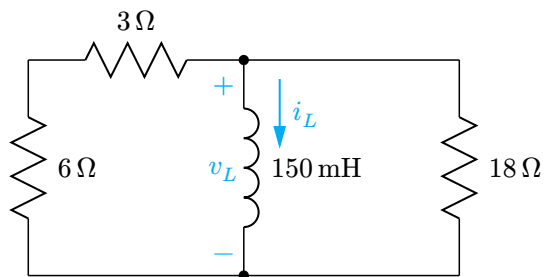
Lab 2: Multi-Switching First-Order Circuits

No. 1

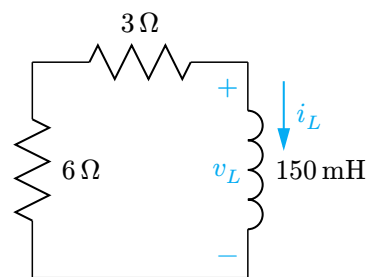
Step 1. Both switches have been closed for a long time. At $t = 0$, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.



$t = 0^-$



$t \in (0, 35) \text{ ms}$



$t = 35 \text{ ms}^+$

for $t = 0^-$:

$$i_s = \frac{60}{4 + (12 \parallel 6 \parallel 3)} = 10.5 \text{ A}$$

$$v_a = v_s - 4i_s = 18 \text{ V}$$

$$\frac{v_a - 60}{4} + \frac{v_a}{12} + \frac{v_a}{6} + i_L = 0$$

$$3v_a + v_a + 2v_a + 12i_L = 180$$

$$\Rightarrow i_L(0^-) = 6 \text{ A}$$

for $t \in [0, 35)$:

$$i_L(0^+) = i_L(0^-) = 6 \text{ A}$$

$$R_{\text{eq}} = (6 + 3) \parallel 18 = 6 \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{150 \text{ mH}}{6 \Omega} = \frac{1}{40} \text{ s}$$

$$i_L(t) = \cancel{i_L(\infty)}^0 + [i_L(0^+) - \cancel{i_L(\infty)}^0] e^{-t/\tau}$$

$$i_L(t) = 6e^{-40t} \text{ for } t \in [0, 35) \text{ ms}$$

$$v_L(t) = L \frac{di_L}{dt}$$

$$\Rightarrow \boxed{v_L(t) = -36e^{-40t} \text{ V}} \text{ for } t \in [0, 35) \text{ ms}$$

for $t \geq 35 \text{ ms}$:

$$i_L(35 \text{ ms}^+) = i_L(35 \text{ ms}^-) = 6e^{-1.4} \text{ A}$$

$$R_{\text{eq}} = 3 + 6 = 9 \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{150 \text{ mH}}{9 \Omega} = \frac{1}{60} \text{ s}$$

$$i_L(t) = \overset{0}{\cancel{i_L(\infty)}} + [i_L(35 \text{ ms}^+) - \overset{0}{\cancel{i_L(\infty)}}] e^{-t/\tau}$$

$$i_L(t) = 6e^{-1.4} e^{-60t}$$

$$i_L(t) = 6e^{-60t-1.4} \text{ A for } t \geq 35 \text{ ms}$$

$$v_L(t) = L \frac{di_L}{dt}$$

$$\Rightarrow \boxed{v_L(t) = -54e^{-60t-1.4} \text{ V}} \text{ for } t \geq 35 \text{ ms}$$

offset $v_L(t)$ for $t \geq 35 \text{ ms}$:

$$v_L(t) = -54e^{-60(t-0.035)-1.4}$$

$$= -54e^{-60t+0.7}$$

thus, for $t \geq 0$:

$$\boxed{v_L(t) = \begin{cases} -36e^{-40t} & \text{V for } t \in [0, 35) \text{ ms} \\ -54e^{-60t+0.7} & \text{V for } t \geq 35 \text{ ms} \end{cases}}$$

Step 2. Simulate the circuit and confirm the results in Step 1.

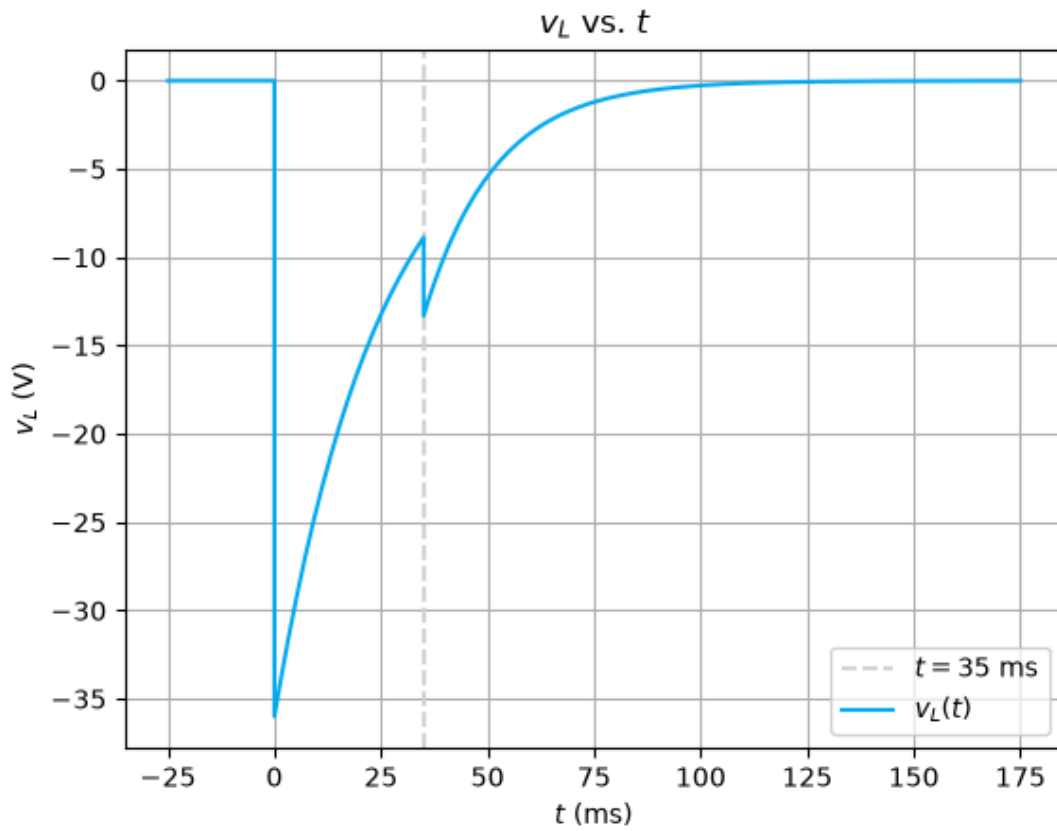


Figure 1: The simulated graph for $t \in [0, 35)$ ms closely matches the theoretical function for $v_L(t) = -36e^{-40t}$ V, while $v_L(t) = -54e^{-60t-1.4}$ V matches it for $t \geq 35$ ms.